

90 Active-Recall Exercises

Fill-in / Calculation / Match / Order — SIKD over Tropical Satellite FSO

Companion to the 360-question bank; same computed answers as the web app.

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How to use

Work each section with pen and paper (calculations) or from memory (fill/match/order), then check Part B. Every item lists its **theory framework** so you memorise the reasoning, not just the answer. Numeric answers were computed from the v3 formulas, identical to the website tatacattn.github.io/for-ttn-USTH/Thesis-M1.

Contents

Part A — Questions	2
S1. Channel physics — calculate	2
S2. Geometry & orbits — calculate	3
S3. Weather & availability	4
S4. Key numbers & terms — fill in	4
S5. Committee & literature — match	5
S6. Reasoning, derivations & defense answers	7
Part B — Answer key & frameworks	9

Part A — Questions

S1. Channel physics — calculate

- E1. [Calculate]** Beam waist $w_0 = \lambda / (2 \cdot \theta_C)$, $\lambda = 1550$ nm, $\theta_C = 10$ urad. Compute w_0 in metres.
Answer: _____ m
- E2. [Calculate]** Spherical slant range at elevation 30 deg ($\zeta = 60$), $H_S = 550$ km, $R_E = 6371$ km. Give L in km.
Answer: _____ km
- E3. [Calculate]** Slant range at elevation 90 deg (satellite at zenith). Give L in km.
Answer: _____ km
- E4. [Calculate]** Beam radius w_L at the receiver for elevation 45 deg. Give w_L in metres.
Answer: _____ m
- E5. [Calculate]** Corrected normalized aperture $\nu_R = \sqrt{(\pi/2) \cdot a_R / w_L}$ at elevation 90 deg ($a_R = 5$ cm). Give ν_R .
Answer: _____ (dimensionless)
- E6. [Calculate]** Geometric loss h_g at elevation 45 deg, in dB (v_3 corrected).
Answer: _____ dB
- E7. [Calculate]** Clear-air extinction σ_{clear} at $V = 10$ km, 1550 nm (Kruse/Kim, $q = 1.3$). Give km^{-1} .
Answer: _____ km^{-1}
- E8. [Calculate]** Rain extinction σ_{rain} at $R = 30$ mm/h (tropical Kone 2024: $\beta = 0.509 R^{0.63}$, $\sigma = \beta / 4.343$).
Answer: _____ km^{-1}
- E9. [Calculate]** Atmospheric transmission h_l in dB at $\zeta = 30$ deg, $V = 10$ km, clear ($R = 0$). $L_{\text{atm}} = H_{\text{atm}} / \cos(\zeta)$.
Answer: _____ dB
- E10. [Calculate]** h_l in dB at $\zeta = 30$ deg, $V = 10$ km, WITH heavy rain $R = 30$ mm/h. Show how much rain adds.
Answer: _____ dB
- E11. [Calculate]** Atmospheric slant path L_{atm} at $\zeta = 60$ deg (elevation 30), $H_{\text{atm}} = 20$ km. Give km.
Answer: _____ km
- E12. [Calculate]** Binary entropy $H_2(0.11)$. Give the value (bits).
Answer: _____ bits
- E13. [Calculate]** Normalized SKR = $\text{Psift} \cdot [1 - 2 \cdot H_2(\text{QBER})]$ at $\text{Psift} = 0.5$, $\text{QBER} = 0.0002$ (v_3 , $f_{\text{EC}} = 1$, $\chi_E = 0$). Give bit/s/Hz.
Answer: _____ bit/s/Hz
- E14. [Calculate]** Absolute SKR (Mbps) = $\text{SKR}_{\text{norm}} \cdot R_b$, with $\text{SKR}_{\text{norm}} = 0.497$ and $R_b = 1$ Gbps. Give Mbps.
Answer: _____ Mbps
- E15. [Calculate]** Shot-noise DC signal power $P = 0.5 \cdot P_T \cdot h_g \cdot h_l$ (v_3 , $\text{NO}_m \text{ K}$), $P_T = 1$ W, $h_g = 3.9 \times 10^{-4}$, $h_l = 0.0954$. Give W.
Answer: _____ W

E16. [Calculate] Cross-talk suppression factor $10^{(-\text{Iso}/10)}$ at $\text{Iso}=15$ dB. Give the linear factor.

Answer: _____ (linear)

E17. [Calculate] Rytov scaling: $\sec^{(11/6)}(60 \text{ deg}) / \sec^{(11/6)}(30 \text{ deg})$. Give the ratio.

Answer: _____ (ratio)

E18. [Calculate] Convert $h_g=2.6e-4$ (linear) to dB.

Answer: _____ dB

S2. Geometry & orbits — calculate

E19. [Calculate] Ground-coverage radius R_{cov} at 30 deg mask, $H_S=550$ km. Give km.

Answer: _____ km

E20. [Calculate] Ground-coverage radius at the 40 deg security mask. Give km.

Answer: _____ km

E21. [Calculate] Max station separation for a DUAL (shared-pass) city pair at 30 deg mask $= 2 \cdot R_{\text{cov}}$. Give km.

Answer: _____ km

E22. [Calculate] Max DUAL separation at the 40 deg mask $= 2 \cdot R_{\text{cov}}(40)$. Give km.

Answer: _____ km

E23. [Calculate] Off-nadir angle η at the satellite for elevation 30 deg. $\sin \eta = R_E \cos e / (R_E + h)$. Give deg.

Answer: _____ deg

E24. [Calculate] Off-nadir angle at elevation 90 deg (zenith). Give deg.

Answer: _____ deg

E25. [Calculate] Great-circle distance Hanoi (21.03N,105.85E) to HCMC (10.82N,106.63E) by haversine. Give km.

Answer: _____ km

E26. [Calculate] Is Hanoi-HCMC (dist above) DUAL at 30 deg mask? Enter the max DUAL separation it is compared against (km).

Answer: _____ km

E27. [Calculate] Elevation angle if zenith angle $\zeta=55$ deg. Give deg.

Answer: _____ deg

E28. [Calculate] Pass frequency per day if 125 passes are seen in a 2-hour window. $\text{freq} = \text{count} \cdot (24/\text{hours})$.

Answer: _____ passes/day

E29. [Calculate] Number of unique city pairs among 8 ground stations $= C(8,2)$. Give the count.

Answer: _____ pairs

E30. [Calculate] Footprint area reduction going 30→40 deg mask: $1 - (R_{\text{cov}40}/R_{\text{cov}30})^2$, as a percentage.

Answer: _____ %

S3. Weather & availability

- E31. [Calculate]** Link availability A at Kuala Lumpur, November ($P_{\text{cloud}}=0.893$). $A=1-P_{\text{cloud}}$.
 Answer: _____ (fraction)
- E32. [Calculate]** Link availability at Bangkok, February ($P_{\text{cloud}}=0.035$). Give fraction.
 Answer: _____ (fraction)
- E33. [Calculate]** Hourly rain rate from $R_{\text{month}}=313.2$ mm, $f_{\text{rain}}=0.15$. $R=R_{\text{month}}/(30*24*f_{\text{rain}})$. mm/h.
 Answer: _____ mm/h
- E34. [Calculate]** Same conversion with the real $f_{\text{rain}}=0.30$ instead of 0.15. mm/h.
 Answer: _____ mm/h
- E35. [Calculate]** $p_{\text{rain}}=\min(f_{\text{rain}}, 1-P_{\text{cloud}})$ with $f_{\text{rain}}=0.30$, $P_{\text{cloud}}=0.893$. Give p_{rain} .
 Answer: _____ (fraction)
- E36. [Calculate]** $p_{\text{clear}}=\max(0, 1-P_{\text{cloud}}-p_{\text{rain}})$ with $P_{\text{cloud}}=0.5$, $f_{\text{rain}}=0.30$. Give p_{clear} .
 Answer: _____ (fraction)
- E37. [Calculate]** Effective SKR = $p_{\text{clear}}*SKR_{\text{clear}} + p_{\text{rain}}*SKR_{\text{rain}}$, $p_{\text{clear}}=0.20$, $SKR_{\text{clear}}=61$, $p_{\text{rain}}=0.30$, $SKR_{\text{rain}}=5$ (Mbps).
 Answer: _____ Mbps
- E38. [Fill in]** What ERA5 cloud-cover percentage threshold defines a 'cloudy hour' (link outage)? Give the number with % sign.
 Answer: _____ %
- E39. [Fill in]** How many years of ERA5/ERA5-Land reanalysis calibrate the weather model? Give the number.
 Answer: _____ years
- E40. [Fill in]** In which local-hour window is the combined cloud+rain 'best FSO window'? (e.g. 05-10h)
 Answer: _____ local
- E41. [Fill in]** Which weather variable (not rain) governs tropical optical availability?
 Answer: _____
- E42. [Fill in]** The tropical rain-attenuation law ($\alpha=0.509$, $\rho=0.63$) was measured in which city?
 Answer: _____

S4. Key numbers & terms — fill in

- E43. [Fill in]** v3 secret key rate at 45 deg elevation (give value with unit, e.g. '61.4 Mbps').
 Answer: _____
- E44. [Fill in]** v3 matching gain (ALG-1 vs ALG-0) in the dry season, as a percentage (e.g. +2.47%).
 Answer: _____
- E45. [Fill in]** Sign of the weather-information gain (aware vs blind): positive or negative?
 Answer: _____

E46. [Fill in] Fraction of days with at least one starved city pair (range, e.g. 91.6-96.5%).

Answer: _____

E47. [Fill in] Coverage radius at the 30 deg mask (km).

Answer: _____

E48. [Fill in] Coverage radius at the 40 deg mask (km).

Answer: _____

E49. [Fill in] How many DUAL city pairs (of 28) can share one satellite pass?

Answer: _____ pairs

E50. [Fill in] Number of real Starlink Shell-1 satellites used.

Answer: _____ satellites

E51. [Fill in] Shell-1 orbital inclination (deg).

Answer: _____ deg

E52. [Fill in] Satellite altitude assumed throughout (km).

Answer: _____ km

E53. [Fill in] QKD carrier wavelength (nm).

Answer: _____ nm

E54. [Fill in] Corrected key modulation index m_K used in the thesis.

Answer: _____

E55. [Fill in] Filter isolation Iso between the key and data subcarriers (dB).

Answer: _____

E56. [Fill in] Real acquisition/handover cost charged per handover (seconds).

Answer: _____

E57. [Fill in] BB84 QBER security threshold above which no secret key can be distilled (%).

Answer: _____

S5. Committee & literature — match

E58. [Match] Match each external examiner to their home research field.

Left: Guillaume Patanchon (APC); Stephane Jacquemoud (IPGP); Cyrille Rosset (IN2P3/APC)

Right (shuffled): CMB detector noise & component separation; Radiative transfer & remote sensing (PROSPECT); Statistical significance (Planck isotropy)

E59. [Match] Match each examiner to the bridge phrase you should use with them.

Left: Jacquemoud; Patanchon; Rosset

Right (shuffled): 'Beer-Lambert = atmospheric correction / radiative transfer'; 'detector noise budget & cross-talk = component separation'; 'paired significance on a full enumeration + block bootstrap'

E60. [Match] Match each examiner to their toughest predicted question.

Left: Jacquemoud; Patanchon; Rosset

Right (shuffled): Is +2.5% significant or within day-to-day scatter?; Why does cross-talk dominate when the link is strong?; Why is Beer-Lambert enough without multiple scattering?

E61. [Match] Match each reference to its role in the thesis.

Left: Vu et al. 2022 (vu2022dtdd); Kone et al. 2024 (kone2024tropical); Polnik et al. 2020 (polnik2020scheduling); Liao et al. 2017 (liao2017satellite)

Right (shuffled): Micius: first satellite QKD from LEO; Nearest scheduling precedent; Noise model & dual-threshold (BBM92 provenance); Tropical rain-attenuation calibration

E62. [Match] Match each reference to its role (set 2).

Left: Andrews & Phillips 2005; Leverrier 2015; Gruneisen et al. 2015; Open-Meteo / ERA5 (openmeteo2023)

Right (shuffled): 10-year real weather climatology; Composable CV-QKD proof (future work); Daytime sky-access for QKD downlink; Rytov variance & H-V turbulence profile

E63. [Match] Match each reference to its role (set 3 — SWaP, PAT & NTN).

Left: Schieler et al. 2023 (TBIRD); Gerard et al. 2025 (PAT); Farid & Hranilovic 2007; 3GPP TR 38.821

Right (shuffled): Non-terrestrial network (NTN) standardisation; Pointing-error term omitted (hg = lower bound); Pointing/Acquisition/Tracking delay → 30 s handover; Small-aperture SWaP comparison (0.4 m RX)

E64. [Match] Match each reference to its role (set 4 — QKD landscape).

Left: Bennett & Brassard 1984; Scarani et al. 2009; Li et al. 2025 (Jinan-1); Toka et al. 2025

Right (shuffled): BB84 protocol (the SKR rate's basis); Microsatellite real-time QKD (cheap sats); Security of practical QKD; Weather-aware LEO routing (classical data)

E65. [Match] Match each examiner to what they will most likely NOT probe deeply.

Left: Son Tong Si (USTH generalist); Jacquemoud (remote sensing); Rosset (statistics)

Right (shuffled): Beam-optics derivations; Deep QKD math (asks motivation/big picture); The Hungarian algorithm internals

E66. [Fill in] Which examiner did the internal supervisor name as the likely primary reviewer? (surname)

Answer: _____

E67. [Fill in] Which classic model is Jacquemoud the author of? (acronym)

Answer: _____

E68. [Fill in] Which satellite mission is the canonical 'first satellite QKD from LEO' you cite in the motivation?

Answer: _____

E69. [Fill in] How many real references are catalogued in the literature review?

Answer: _____ references

E70. [Fill in] In one word, the strategy with this non-QKD panel: report results with ____ (integrity vs. spin).

Answer: _____

E71. [Fill in] Complete Jacquemoud's answer: 'I treat cloud as a ____ outage, so multiple scattering only refines an already-lost state.'

Answer: _____

E72. [Fill in] Complete Rosset's answer: 'It is not a random sample — I ____ every day in two 10-year proxy months.'

Answer: _____

E88. [Fill in] Vu et al. 2022 implements which protocol (not BB84)? Give the acronym.

Answer: _____

E89. [Order] Order the four v3 fixes by the stage they touch (channel $\rightarrow$ receiver).

Items (shuffled): Rytov 0.56 $\rightarrow$ 2.25; cross-talk $/2\rightarrow/8$; erf coefficient $\sqrt{(2)}\rightarrow\sqrt{(\pi/2)}$; spherical slant range

E90. [Fill in] The daytime-background penalty is described as an 'order-of-____ estimate', not a proof. (one word)

Answer: _____

Part B — Answer key & theory frameworks

S1. Channel physics — calculate

E1. 0.0775 m (tol 0%)

$$w_0 = 1550e-9/(2*10e-6) = 0.0775 \text{ m.}$$

Framework: $w_0 = \lambda/(2*\theta_C)$; tighter divergence $\rightarrow$ larger waist $\rightarrow$ slower spreading.

E2. 992.778 km (tol 1%)

$$\epsilon = 30: L = 992.8 \text{ km (NOT the flat-Earth } H_S/\cos(\zeta) = 1100 \text{ km).}$$

Framework: $L = \sqrt{(R_E^2 \sin^2 \epsilon + 2 R_E H_S + H_S^2)} - R_E \sin \epsilon$; corrected from flat-Earth 1100 km.

E3. 550 km (tol 0%)

$$\epsilon = 90: L = H_S = 550 \text{ km.}$$

Framework: At zenith the slant range equals the satellite altitude.

E4. 4.7696 m (tol 3%)

$$L(45) = 749 \text{ km} \rightarrow wL = 4.77 \text{ m.}$$

Framework: $wL = w_0 \sqrt{1 + (L*\lambda/(\pi*w_0^2))^2}$; far-field grows $\sim$ linearly with L .

E5. 0.017893 (dimensionless) (tol 3%)

$$\nu_R = 1.2533*0.05/3.502 = 0.0179.$$

Framework: ν_R uses $\sqrt{\pi/2} = 1.2533$ (v3 fix), NOT $\sqrt{2}$; old value made hg optimistic by 1.05 dB.

E6. -36.5804 dB (tol 0.02 abs)

$$hg(45 \text{ deg elev}) = -36.6 \text{ dB.}$$

Framework: $hg = [\text{erf}(\nu_R)]^2$; $hg \sim 1/L^2$ since $\nu_R \ll 1$; dominant link-budget term (>30 dB).

E7. 0.101728 km⁻¹ (tol 2%)

$$q(10) = 1.3 \rightarrow \sigma = 0.1017 \text{ km}^{-1}.$$

Framework: $\sigma = (3.912/V)(\lambda/550)^{-q(V)}$; longer wavelength scatters less.

E8. 0.998882 km⁻¹ (tol 2%)

$$\beta = 0.509*30^{0.63} = 4.34 \text{ dB/km; } \sigma = 0.999 \text{ km}^{-1}.$$

Framework: Power-law $\rho = 0.63 < 1 \rightarrow$ sub-linear; $4.343 = 10/\ln 10$ converts dB $\rightarrow$ Nepers.

E9. -10.2029 dB (tol 0.02 abs)

$$L_{\text{atm}} = 23.09 \text{ km; } hl = \exp(-0.1017*23.09) = 0.0954 \rightarrow -10.2 \text{ dB.}$$

Framework: $hl = \exp(-\sigma_{\text{total}}*L_{\text{atm}})$; $L_{\text{atm}} = (H_{\text{atm}})/\cos(\zeta)$, $H_{\text{atm}} = 20 \text{ km}$.

E10. -110.387 dB (tol 0.03 abs)

$$\sigma_{\text{total}} = 1.101; hl = \exp(-1.101*23.09) \rightarrow \sim -110 \text{ dB.}$$

Framework: Rain adds ~ 100 dB vs clear $\rightarrow$ near link cut; why cloud/heavy-rain $\sim$ outage.

E11. 40 km (tol 1%)

$$L_{\text{atm}} = 20/\cos 60 = 40 \text{ km (double the zenith value).}$$

Framework: $L_{\text{atm}} = (H_{\text{atm}} - H_U)/\cos(\zeta)$; low passes traverse more atmosphere ($1/\cos$).

E12. 0.499916 bits (tol 1%)

$$H_2(0.11) = 0.4999.$$

Framework: $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$; $H_2(0.11) = 0.4999 \rightarrow$ why SKR=0 near QBER 11%.

E13. 0.497254 bit/s/Hz (tol 3%)

$$H_2(2e-4) \sim 0.0027; 1 - 2*0.0027 = 0.9946; *0.5 = 0.497.$$

Framework: $\text{SKR}_{\text{norm}} = \text{Psift}[1 - (1 + f_{\text{EC}})H_2(Q) - \chi E]$; EC-leakage term makes it hit 0 at $Q \sim 11\%$.

E14. 497 Mbps (tol 3%)

$$0.497*1000 = 497 \text{ Mbps (illustrative; table value 61.4 uses the real Psift/QBER).}$$

Framework: $\text{SKR}_{\text{bps}} = \text{SKR}_{\text{norm}}*R_b$; v3 @45 deg ~ 61 Mbps.

E15. 1.9e-05 W (tol 3%)

$$0.5*1*3.9e-4*0.0954 = 1.86e-5 \text{ W.}$$

Framework: v3 fix: shot noise uses total DC photocurrent, not the m_K -scaled key power (was 20x too small).

E16. 0.031623 (linear) (tol 2%)

$$10^{(-1.5)} = 0.0316.$$

Framework: $\sigma_{CT}^2 \sim 10^{(-\text{Iso}/10)/8}$; better isolation cuts the dominant noise.

E17. 2.73754 (ratio) (tol 2%)

$$(2/1.1547)^{(11/6)} = 2.74.$$

Framework: $\sigma_R^2 \sim \sec^{(11/6)}(\text{zeta})$; turbulence worsens at low elevation.

E18. -35.8503 dB (tol 0.02 abs)

$$10 \log_{10}(2.6e-4) = -35.8 \text{ dB}.$$

Framework: $\text{dB} = 10 \log_{10}(\text{linear})$; geometric loss ~ -36 dB.

S2. Geometry & orbits — calculate

E19. 793.497 km (tol 2%)

$$R_{\text{cov}}(30 \text{ deg}) = 793 \text{ km}.$$

Framework: $R_{\text{cov}} = R_E \psi$, $\psi = \arccos(R_E/(R_E+h) \cos e) - e$; the 950 km 'radius' in old paper was slant range.

E20. 573.417 km (tol 2%)

$$R_{\text{cov}}(40 \text{ deg}) = 573 \text{ km}.$$

Framework: Stricter mask $\rightarrow$ smaller footprint ($\sim 28\%$ loss) but safer QBER.

E21. 1586.99 km (tol 2%)

$$2 * 793 = 1587 \text{ km}.$$

Framework: classify_pair : DUAL if distance $\leq 2 * R_{\text{cov}}$; the 14/14 split threshold.

E22. 1146.83 km (tol 2%)

$$2 * 573 = 1146 \text{ km}.$$

Framework: Stricter mask shrinks the DUAL threshold too.

E23. 52.8639 deg (tol 2%)

$$\sin \eta = 6371 * \cos 30 / 6921 = 0.797 \rightarrow \eta = 52.8 \text{ deg}.$$

Framework: Zero at zenith, grows as elevation drops; the transmit-side pointing angle.

E24. 0 deg (tol 0.01 abs)

$$\eta = 0 \text{ deg}.$$

Framework: At the station's zenith the line of sight IS the nadir direction.

E25. 1138.35 km (tol 3%)

$$\sim 1140 \text{ km}.$$

Framework: haversine is symmetric; DUAL/SF classification uses this distance.

E26. 1586.99 km (tol 2%)

$$\text{Threshold } 1587 \text{ km}; 1140 < 1587 \rightarrow \text{DUAL}.$$

Framework: Compare distance (~ 1140 km) to $2 * R_{\text{cov}}(30) = 1587$ km $\rightarrow$ DUAL ($1140 < 1587$).

E27. 35 deg (tol 0.001 abs)

$$90 - 55 = 35 \text{ deg}.$$

Framework: elevation = 90 - zenith; the two complementary receive-side angles.

E28. 1500 passes/day (tol 1%)

$$125 * 12 = 1500 \text{ passes/day}.$$

Framework: $\text{pass_frequency_per_day}$ extrapolates the enumerated window.

E29. 28 pairs (tol 0.001 abs)

$$8 * 7/2 = 28.$$

Framework: 28 pairs to serve with 8 stations $\rightarrow$ structural starvation.

E30. 47.7783 % (tol 5%)

$$1 - (573/793)^2 = \sim 48\% \text{ area (about } 28\% \text{ radius)}.$$

Framework: Area $\sim \text{radius}^2$; the security mask's coverage cost.

S3. Weather & availability

E31. 0.107 (fraction) (tol 2%)

$A=1-0.893=0.107$ (10.7%).

Framework: $A=p_{\text{clear}}+p_{\text{rain}}=1-P_{\text{cloud}}$; invariant to rain fraction; worst cell in the dataset.

E32. 0.965 (fraction) (tol 1%)

$A=0.965$ (96.5%).

Framework: Best cell; $A=1-P_{\text{cloud}}=0.965$; driven by cloud, not rain.

E33. 2.9 mm/h (tol 2%)

$313.2/108 = 2.90$ mm/h.

Framework: Tropical rain is intermittent; intensity concentrates in raining hours.

E34. 1.45 mm/h (tol 2%)

$313.2/216 = 1.45$ mm/h.

Framework: Larger f_{rain} spreads rain over more hours → lower instantaneous intensity.

E35. 0.107 (fraction) (tol 2%)

$\min(0.30, 0.107)=0.107$.

Framework: When cloud is very high, p_{rain} is capped by $(1-P_{\text{cloud}})$.

E36. 0.2 (fraction) (tol 2%)

$p_{\text{rain}}=0.30$; $p_{\text{clear}}=1-0.5-0.30=0.20$.

Framework: Three-state split; clear gets what's left after cloud and rain.

E37. 13.7 Mbps (tol 2%)

$0.20*61+0.30*5 = 13.7$ Mbps.

Framework: Cloud state contributes 0; weighted average over clear+rain.

E38. 85% %

Framework: CLOUD_OUTAGE_THRESHOLD=85%; a modelling choice with a sensitivity analysis.

E39. 10 years

Framework: 2015-2024; long record captures interannual variability.

E40. 05-10h local

Framework: Cloud lowest mid-morning, rain peaks afternoon → best window ~05-10h, in DAYLIGHT (day/night tension).

E41. cloud

Framework: Cloud is total optical blockage; rain only attenuates. The core meteorological finding.

E42. Abidjan

Framework: Kone et al. 2024; real tropical FSO measurements, defensible under a remote-sensing examiner.

S4. Key numbers & terms — fill in

E43. 61.4 Mbps

Framework: Table 6.1 v3 at $m_K=0.15$; up from pre-v3 13.3 Mbps after the /8 cross-talk fix.

E44. +2.47%

Framework: Modest but real coordination gain over 620 enumerated days.

E45. negative

Framework: -0.29%/-0.23%: weather-awareness re-assigns more → handovers outweigh the benefit.

E46. 91.6-96.5%

Framework: Structural (8 stations / 28 pairs), unchanged across every correction round.

E47. 793 km

Framework: Ground-track radius (NOT the 950 km slant range of the old paper).

E48. 573 km

Framework: ~28% smaller footprint than 30 deg, for a safer QBER margin.

E49. 14 pairs

Framework: The other 14 are store-and-forward; a clean geographic split.

E50. 1019 satellites

Framework: Filtered by inclination 53.15-53.22 deg; real, reproducible TLEs.

E51. 53 deg

Framework: Covers ~95% of world population incl. all ASEAN; launch-efficient from Cape Canaveral.

E52. 550 km

Framework: The channel model is parameterised for this single altitude.

E53. 1550 nm

Framework: Telecom C-band: eye-safe, low loss, mature components, weak aerosol scattering.

E54. 0.15

Framework: Reset from 0.05 to the source manuscript's feasible value; drops QBER below 10^{-3} .

E55. 15 dB

Framework: Finite isolation lets the data channel leak in as cross-talk.

E56. 30 s

Framework: Every re-assignment loses 30 s of delivered key (v3).

E57. 11%

Framework: $1-2 \cdot H_2(Q)=0$ at $Q \sim 0.11$; the Shor-Preskill threshold.

S5. Committee & literature — match

E58. Guillaume Patanchon (APC) → CMB detector noise & component separation; Stephane Jacquemoud (IPGP) → Radiative transfer & remote sensing (PROSPECT); Cyrille Rosset (IN2P3/APC) → Statistical significance (Planck isotropy)

Framework: Bridge each part of the thesis to the examiner's own field; they are space-science generalists, not QKD specialists.

E59. Jacquemoud → 'Beer-Lambert = atmospheric correction / radiative transfer'; Patanchon → 'detector noise budget & cross-talk = component separation'; Rosset → 'paired significance on a full enumeration + block bootstrap'

Framework: Speak each examiner's mother tongue; do not assume QKD fluency.

E60. Jacquemoud → Why is Beer-Lambert enough without multiple scattering?; Patanchon → Why does cross-talk dominate when the link is strong?; Rosset → Is +2.5% significant or within day-to-day scatter?

Framework: The three trap questions; master the MECHANISM, not just the number.

E61. Vu et al. 2022 (vu2022dtdd) → Noise model & dual-threshold (BBM92 provenance); Kone et al. 2024 (kone2024tropical) → Tropical rain-attenuation calibration; Polnik et al. 2020 (polnik2020scheduling) → Nearest scheduling precedent; Liao et al. 2017 (liao2017satellite) → Micius: first satellite QKD from LEO

Framework: Each citation anchors one layer of the work; know which paper does what.

E62. Andrews & Phillips 2005 → Rytov variance & H-V turbulence profile; Leverrier 2015 → Composable CV-QKD proof (future work); Gruneisen et al. 2015 → Daytime sky-access for QKD downlink; Open-Meteo / ERA5 (openmeteo2023) → 10-year real weather climatology

Framework: The physics, security, daytime, and weather anchors respectively.

E63. Schieler et al. 2023 (TBIRD) → Small-aperture SWaP comparison (0.4 m RX); Gerard et al. 2025 (PAT) → Pointing/Acquisition/Tracking delay → 30 s handover; Farid & Hranilovic 2007 → Pointing-error term omitted (hg = lower bound); 3GPP TR 38.821 → Non-terrestrial network (NTN) standardisation

Framework: The engineering/context anchors; each justifies one modelling choice or positioning.

E64. Bennett & Brassard 1984 → BB84 protocol (the SKR rate's basis); Scarani et al. 2009 → Security of practical QKD; Li et al. 2025 (Jinan-1) → Microsatellite real-time QKD (cheap sats); Toka et al. 2025 → Weather-aware LEO routing (classical data)

Framework: Foundations, security framework, and the two nearest 'cheap-sats' / 'weather-aware' precedents.

E65. Son Tong Si (USTH generalist) → Deep QKD math (asks motivation/big picture); Jacquemoud (remote sensing) → The Hungarian algorithm internals; Rosset (statistics) → Beam-optics derivations
 Framework: Audience analysis: aim answers at each examiner's strengths, keep others high-level.

E66. Jacquemoud

Framework: IPGP radiative-transfer expert; will scrutinise the atmospheric model most.

E67. PROSPECT

Framework: Leaf radiative-transfer model 400-2500 nm; his home ground is extinction/attenuation.

E68. Micus

Framework: Liao et al. 2017, Nature 549:43; the existence proof QKD works from orbit.

E69. 34 references

Framework: Built directly from the thesis references.bib, mapped to role + examiner.

E70. integrity

Framework: Flag limitations proactively; the maturity signal that wins a science panel.

E71. binary

Framework: Concede Beer-Lambert is first-order; justify cloud=outage → no multiple scattering needed.

E72. enumerate

Framework: Full enumeration (620 days) → no sampling error; paired day-by-day test isolates the ~2.5% effect.

S6. Reasoning, derivations & defense answers

E73. channel (hg, hl, σ_X^2) → noise budget (4 terms) → dual-threshold (d0,d1) → Psift & QBER (Gauss-Hermite) → SKR (Shor-PreSkill)

Framework: compute_sikd_performance wrapper: each stage feeds the next; SKR is the end product.

E74. Transmitted power P_T → Geometric spreading h_g → Atmospheric attenuation h_l → Turbulence fading h_a → Received power

Framework: $h = h_g * h_l * h_a$; three independent multiplicative loss mechanisms.

E75. Clear (attenuation only) → Rain (added attenuation) → Cloud (link OFF)

Framework: Three-state model; cloud is binary outage, not gradual attenuation.

E76. Station-satellite matching (Hungarian) → Build matched pass table → Pair-key allocation (greedy/ILP/maxmin) → $K_q = \min(\text{sideA}, \text{sideB})$

Framework: Level 1 matching → level 2 allocation on a shared trusted-relay satellite.

E77. Thermal noise (floor, weak link) → Shot noise (grows with power) → Cross-talk ($\sim P^2$, strong link)

Framework: Cross-talk $\sim$ received power squared → dominates exactly when the link is strongest.

E78. $-\sigma_X^2$

Framework: $E[e^{2X}] = \exp(2\mu_X + 2\sigma_X^2) = 1$ requires $\mu_X = -\sigma_X^2$ (unit-mean multiplicative fading).

E79. 11%

Framework: $1 - 2H_2(Q) = 0$ at $Q \sim 0.11$; the term added in the v3 SKR fix restores this.

E80. 2

Framework: $\sigma_{CT}^2 \sim (h_g h_l)^2 \rightarrow P^2$; the counter-intuitive SIKD result.

E81. 8

Framework: Photocurrent's leading $1/2$ makes amplitude $(1/2)(\dots)$; variance $A^2/2 \rightarrow /8$.

E82. min

Framework: $K_q = \min(\text{sideA}, \text{sideB})$; one weak link caps the improvement → structural ceiling.

E83. handovers

Framework: Measured mechanism (I counted the handovers directly), not a bug; small, same sign both seasons.

E84. capacity

Framework: 8 stations cannot serve 28 pairs; the fix is more infrastructure.

E85. Hungarian

Framework: `match_weather_aware` uses `scipy linear_sum_assignment` on the negated weight matrix.

E86. trusted

Framework: The relay learns both keys; security holds only if the relay itself is honest.

E87. network

Framework: The central claim; a shift in the unit of evaluation.

E88. BBM92

Framework: The provenance caveat: the 11% BB84 bound is a borrowed, openly-stated assumption.

E89. erf coefficient $\sqrt{2} \rightarrow \sqrt{\pi/2} \rightarrow$ spherical slant range $\rightarrow$ Rytov $0.56 \rightarrow 2.25 \rightarrow$ cross-talk $/2 \rightarrow /8$

Framework: Three geometric/turbulence fixes in the channel, then the receiver cross-talk fix (largest impact).

E90. magnitude

Framework: Honest confidence labelling; a plausibility bound from an FOV sweep, not a full simulation.